

Top K From Sorted Structures

“ Concepts & Qns ”



Video - 1



codestorywithmik



Cs with MIK



codestorywithMIK

Motivation :

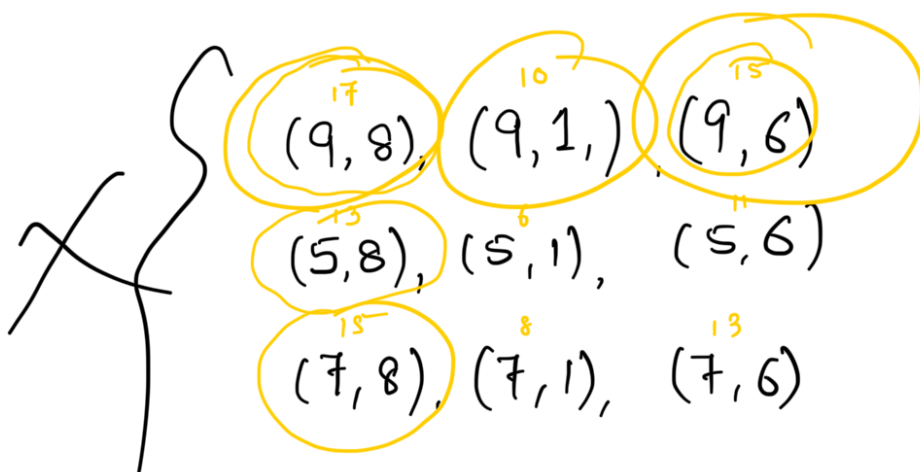
Optimization:

$$\text{Growth} \propto \frac{1}{(\text{Comfort zone})}$$

MIK

Introduction

$$\begin{array}{l} \text{nums1} = \{9, 5, 7\} \rightarrow \\ \text{nums2} = \{8, 1, 6\} \rightarrow \end{array} \quad \begin{array}{c} \text{nums1, nums2} \\ \left(\begin{array}{cc} \downarrow & \downarrow \end{array} \right)$$



Top 4 pair sums =

(9, 8) (9, 6)

(7, 8) (5, 8)

for (i = 0; i < nums1.length(); i++) {

$O(n^2)$

```
for (j=0; j<nums2.length; j++) {
```

```
    pair(nums1[i], nums2[j]);
```

```
    sum = nums1[i] + nums2[j];
```

```
}
```

```
}
```

$$10^5 * 10^5 = 10^{10}$$

T.L.G

$O(n^2)$

(i) Generated every possible pair sum.

(ii) Sorted everything.

(iii) we picked top-K sum.

nums1 = {9, 7, 5}

nums2 = {8, 6, 1}

(9,8), (9,6), (9,1)

17, 15, 10

(7,8), (7,6), (7,1)

15, 13, 8

(5,8), (5,6), (5,1)

13, 11, 6

Ordering.
=

find top k elem

Stream 1 $\rightarrow$ 100, 90, 80, 70

Stream 2 $\rightarrow$ 99, 95, 60, 20

Stream 3 $\rightarrow$ 98, 97, 50, 10

top 6 elements



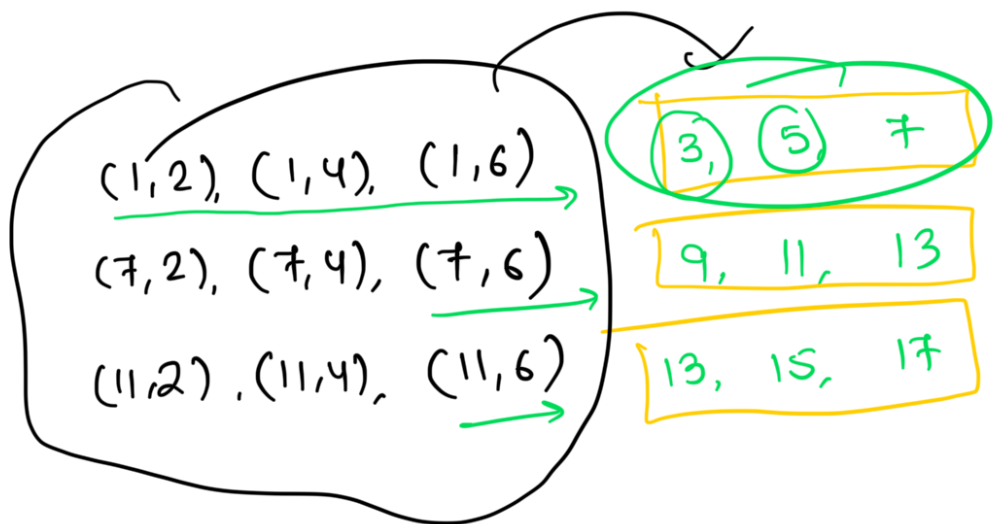
$\downarrow$
sort.

nums1 = {1, 7, 11}

nums2 = {2, 4, 6}

$k=3$

K pairs
with smallest
Sum



Tok K
 Smallest K
 Largest K

$\leadsto$ Heap

$nums1 = \{1, 7, 11\}$
 $nums2 = \{2, 4, 6\}$

$(1,2) \rightarrow 3$

$sum = 7$

$i = 0$

$j = 2$

$k = 3$

K pairs
 with Smallest
 Sum

7 11
 $(1, 6)$ $(7, 4)$
 $\uparrow$ $\uparrow$

$7, (0, 2)$

Heap.

9, (1,0)
11, (1,1)

min-heap.

$$\text{result} = \{(1,2), (1,4), (1,6)\}$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ 3 & 5 & 7 \end{array}$$

{

}

} Top K.

How to identify ???

✓ K^{th} Smallest

✓ TOP K

✓ K^{th} largest

✓ Best K

✓ Merge K sorted lists/arrays etc.

✓ Sorted rows in 2D matrix

→ Find top K

✓ Sorted columns in 2D matrix

→ Find top K

✓ Pair Sums → Find top K / min.

✓ Subarray Sums → Find top K

Check :-

✓ Avoid generating all possibilities

✓ Can Heap help ?

✓ I only care about next best (smallest/largest).

~~Heap~~ →

